TERTIARY INFOTECH ACADEMY PTE LTD LESSON PLAN

Autodesk Certified Professional (ACP) for Revit Mechanical Design

WSQ Course Code: TGS-2023036661

Version v7.0 · 14 September 2026

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Document Version Control Record

Version	Date	Change	Author
6.0	Legacy reference	Original Mechanical slide deck (reference only)	Tertiary Infotech Academy
7.0	14 September 2026	Rebuilt aligned mechanical deck, labs, learner guide, lesson plan and assessment	Tertiary Infotech Academy

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Course information

TGS-2023036661 · WSQ · 4 days / 32 hours · TSC BEV-TDR-4006-1.1-1

- LO1: Execute mechanical calculations and build conceptual duct and piping designs with a Basis of Design.
- LO2: Produce coordinated mechanical schedules, plans, sections, sheets and revisions.
- LO3: Create and validate MEP families, connectors and parameters for accurate BIM models.
- LO4: Review analytical spaces, zones, energy inputs and calculated values against project criteria.
- LO5: Coordinate linked models, clashes, standards and handover evidence efficiently.

Four-day schedule

Day 1 — 8 training hours

Time	Minutes	Content and slide references	Evidence
09:00–10:30	90	Topic mechanism and worked model; Topic 1; slides 10–87	Lab 01–03 RVT/schedule/PDF
10:30–10:45	15	Break; Topic 1; slides 10–87	Lab 01–03 RVT/schedule/PDF
10:45–13:00	135	Guided mechanical modelling and evidence; Topic 1; slides 10–87	Lab 01–03 RVT/schedule/PDF
13:00–14:00	0	Lunch; Topic 1; slides 10–87	Lab 01–03 RVT/schedule/PDF
14:00–15:30	90	Lab application and peer review; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF
15:30–15:45	15	Break; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF
15:45–18:00	135	Lab completion, QA and PM attendance; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF

Training total: 480 minutes including supervised breaks; lunch excluded.

Day 2 — 8 training hours

Time	Minutes	Content and slide references	Evidence
09:00–10:30	90	Topic mechanism and worked model; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF

10:30–10:45	15	Break; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF
10:45–13:00	135	Guided mechanical modelling and evidence; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF
13:00–14:00	0	Lunch; Topic 2; slides 88–165	Lab 04–06 RVT/schedule/PDF
14:00–15:30	90	Lab application and peer review; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
15:30–15:45	15	Break; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
15:45–18:00	135	Lab completion, QA and PM attendance; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF

Training total: 480 minutes including supervised breaks; lunch excluded.

Day 3 — 8 training hours

Time	Minutes	Content and slide references	Evidence
09:00–10:30	90	Topic mechanism and worked model; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
10:30–10:45	15	Break; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
10:45–13:00	135	Guided mechanical modelling and evidence; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
13:00–14:00	0	Lunch; Topic 3; slides 166–243	Lab 07–09 RVT/schedule/PDF
14:00–15:30	90	Lab application and peer review; Topic 4; slides 244–321	Lab 10–12 RVT/schedule/PDF
15:30–15:45	15	Break; Topic 4; slides 244–321	Lab 10–12 RVT/schedule/PDF
15:45–18:00	135	Lab completion, QA and PM attendance; Topic 4; slides 244–321	Lab 10–12 RVT/schedule/PDF

Training total: 480 minutes including supervised breaks; lunch excluded.

Day 4 — 8 training hours

Time	Minutes	Content and slide references	Evidence
09:00–10:30	90	Topic 5: collaboration and model review; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF

10:30–10:45	15	Break; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
10:45–13:00	135	Labs 13–15 final coordination and evidence; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
13:00–14:00	0	Lunch; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
14:00–15:00	60	Written Assessment (WA-SAQ) — 8 questions; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
15:00–15:15	15	Break; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
15:15–17:15	120	Practical Performance (PP) — 3 integrated tasks; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF
17:15–18:00	45	Assessment evidence, sign-off and PM attendance; Topic 5; slides 322–399	Lab 13–15 RVT/schedule/PDF

Training total: 480 minutes including supervised breaks; lunch excluded.

Assessment

WA (SAQ): 60 minutes; Practical Performance: 120 minutes. Brief candidates before the Day 4 assessment window. The remaining Day 4 practice blocks support final model and evidence review.

Alignment

Topic 1 — Modeling for mechanical design: K1 K2 A1 A2; Labs 1–3.

Topic 2 — Documentation for mechanical design: K3 K4 A3 A4; Labs 4–6.

Topic 3 — Families for mechanical design: K5 K6 A5 A6; Labs 7–9.

Topic 4 — Analysis for mechanical design: K7 A7; Labs 10–12.

Topic 5 — Collaboration for mechanical design: K8 A8 A9; Labs 13–15.